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Absence of Correlations between Opaque Petrology and Natural Remanence Polarity in Canary Island Lavas

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Summary

The titanomagnetite oxidation state, separate ilmenite abundance and other petrological properties have been measured in specimens from 168 Miocene to Pliocene Canary Island lava flows. No correlation is found between opaque petrological parameters and natural remanence polarity regardless of the way in which the data for the lava collection is subdivided.

1. Introduction

This paper describes a continuation of investigations into the relationship between natural remanence polarity and opaque petrology in igneous rocks. This series of investigations was first promoted by Blackett who in 1962 suggested that the possibility that non-repeatable self reversal was the mechanism responsible for reversely magnetized rocks could be tested for by searching for petrological differences between normally and reversely magnetized rocks. Several examples of statistical differences all in the same sense have been found (Wilson & Watkins 1967; Ade-Hall & Wilson 1969). In the last few years convincing evidence that at least many relatively young reversely magnetized lavas owe their polarity to eruption during a reversed field has arisen as a result of the accurate and precise potassium–argon dating of Pliocene to recent lavas (with results summarized in Cox, Doell & Dalrymple 1968). Further evidence in support of the frequent occurrence of a reversed state for the geomagnetic field during the Tertiary derives from the world-wide distribution of correlatable oceanic magnetic anomaly patterns (Heirtzler *et al.* 1968). Thus we are faced with the situation that petrological differences occur, at least sometimes, between normally and reversely magnetized lavas, while at the same time, there is convincing evidence that the geomagnetic field can have reverse polarity.

As Bullard (1968) has pointed out, the existence of the observed petrological associations, usually of a feature suggesting higher oxygen fugacity during lava cooling in a reverse polarity field, appears to be one of the major unsolved problems in geophysics, since no satisfactory explanation is as yet forthcoming. Several suggestions have been made in attempts to begin to answer this difficult point. Wilson & Watkins (1967) state that the evidence is such that ' A causal connection, however indirect, between field polarity and oxidation state in lavas would permit the evidence to be reconciled ' (pages 421/422). Again, Watkins & Haggerty (1968) say that ' Until a meaningful model can be proposed, the observations described in this communication can only be *understood* to be a function of sampling ' (page 315).

The reasons for studying the Canary Island lavas from this viewpoint are both to expand the limited amount of combined petrological–magnetic data available and to carry out a test for correlations in the previously untested apparently oceanic geological environment.

2. Field and laboratory methods

(*a*) *Field methods.* Oriented cores have been collected from several hundred extrusive and intrusive igneous rocks in the seven major Canary Islands, as the basis of diverse geological and geophysical studies. Some of the palaeomagnetic results (Watkins, Richardson & Mason 1966) and potassium–argon age measurements (Abdel-Monem, Gast & Watkins 1967) have already been reported.

The purpose of the study described in this paper is to test for the possibility of a relationship between oxidation and polarity in a collection from the sampled lavas and thus it is not necessary to list the fine stratigraphic details of each body sampled. We list below the major stratigraphic units from which the specimens have been taken.

1. Lanzorote. Specimens are restricted to the Basaltic Tableland Series (Hausen, 1959) or Basaltic Series I (Fuster, Santin & Sagredo 1968a), which outcrop on the Guatafay–Farmara Complex in the north of the island, and in the Los Ajaches area in the southwest. Abdel-Monem, Gast & Watkins (1967) have shown that the age of these lavas is from 5 to 12 My.

2. Fuertaventura. Specimens are mainly from the Basaltic Tableland Series (Hausen 1958) or Basaltic Series I (Fuster *et al.* 1968a), which outcrop in the Jandia peninsula and in the central-eastern half of the island. These are Upper Pliocene to Middle Miocene in age (Abdel-Monem, Gast & Watkins 1968).

3. Grand Canary. The specimens are from several stratigraphic units. The oldest (middle Miocene) lavas which outcrop in the northwest and southwest part of the island are termed the Tableland Formation by Hausen (1962) and, again, the Basaltic Series I, by Fuster *et al.* (1968b). A Lower Pliocene unit sampled is the Trachytic–Phonolitic Series of Bourcart & Jeremine (1937) which is termed the Phonolite Series by Fuster *et al.* (1968b) and the Nepheline-bearing Phonolites by Hausen (1962). Specimens were also taken from the Roque Nublo Series (Fuster *et al.* 1968b) or the Vindobodium Basanite (Bourcart & Jeremine 1937), which is Upper Pliocene in age. Finally, several specimens were obtained from the Post–Miocene Basalts (Hausen 1962) which are Plio-Pleistocene in age.

4 Tenerife. The Old Basalt Series (Fuster *et al.* 1968c) of the Anaga peninsula and the Ladera de Guimar provided most of the specimens from this island. Phonolitic rocks occur on Tenerife to a greater extent than on Grand Canary. These were sampled in the Canadas Series (Fuster *et al.* 1968c) or in the Phonolitic Formation of the Pre-Caldera Volcanics (Hausen 1956) on the Tiaga block, and in the Vilaflor area. The Post-Caldera Basalts (Hausen 1956) provided the rest of the specimens.

5. La Gomera Specimens were only taken from the Old Basalts (Bravo 1964) to Middle Miocene in age (Abdel-Monem, Gast & Watkins 1967).
of Lower Pliocene age (Abdel-Monem, Gast & Watkins 1967) and from the younger Horizontal Basalts (Hausen 1965) both of which outcrop on the major road northwest of San Sebastian.

6. La Palma. Only the younger flows which outcrop the eastern, southern, and west-central coasts of the island were sampled. Hausen (personal communication, 1967) calls these lavas the Adventive Alkali Basaltic Volcanics. The sampled units are all Pleistocene in age (Abdel-Monem, Gast & Watkins 1967).

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7. Hierro. Specimens are from the Tableland Formation (Hausen 1964) which, according to Abdel-Monem, Gast & Watkins (1967) is of Pleistocene age. These lavas outcrop west of Valverde, and in the wall of the great El Golfo scarp.

We have divided the specimens into three very broad petrological divisions, which are based purely on field identification: ' phonolite ' (a platey, usually dense, medium to dark grey rock, which sometimes features porphyritic feldspathoids), ' ankaramite ' (a basalt with abundant porphyritic olivine and pyroxene); and ' basalt ' (including olivine basalt, augite basalt, plagioclase basalt, and other varieties, many of which cannot be distinguished in the field).

We have assembled specimens from 168 lavas. No intrusive rocks are used, since some form of standard igneous environment is clearly desirable for the purpose of the experiment described here.

2(*b*) *Laboratory methods.* From each core specimens have been taken for palaeomagnetic analysis and for the manufacture of a polished section.

All palaeomagnetic cores have been demagnetized in alternating magnetic fields up to 300 Oersteds, and magnetic intensities and directions have been measured after each demagnetization step. Reliable polarity data result from measurements of this type being carried out on up to eight specimens per flow.

The oxidation state of the titanomagnetites has been measured using the Reichert Zetopan-pol microscope, using magnifications of $\times 1200$ under oil. The original five-fold *oxidation class* system of Wilson & Watkins (1967), which was replaced by a six-fold *oxidation index* system (Watkins & Haggerty 1968) involves possibly subjective estimates of the overall oxidation state of a given specimen. This was superseded by the more quantitative approach of Wilson *et al.* (1968) in which the number (n) of 300 separate titanomagnetite grains falling in each of the oxidation indices (i) I to VI, corresponding to lowest oxidation homogeneous titanomagnetite to intergrowths of pseudobrookite, hematite, and sometimes rutile was noted, and the magnetic oxidation number, M, computed where

$$M = \sum_{i=1}^{i=6} \left(\frac{n_i i}{n} \right).$$

Ade-Hall *et al.* (1968) have described the most recent refinement of this optical classification system which has been used in this study.

It should be emphasized that the above-mentioned modifications to the petrological analysis scheme do not prevent meaningful comparison between data reported using different methods of analysis.

Briefly, in this study, average within-specimen opaque petrological data were obtained by the classification and dimension measurement of about one hundred opaque grains per specimen. Opaque grains are classified both in terms of their original (immediately post crystallization) type, *i.e.* as titanomagnetite, ilmenite, pyrite or pyrrhotite, and their type and degree of deuteric or later alteration. Two main types of alteration are recognized, titanomagnetite and ilmenite deuteric alteration, which takes place during the initial cooling of the lava, and titanomagnetite granulation, which probably takes place as a result of considerable burial by younger flows. More titanomagnetite than ilmenite deuteric oxidation steps have been recognised and a magnetite deuteric oxidation scale running from 1·00 (oxidation absent) to 6·00 (maximum oxidation) has been established by Wilson *et al.* (1968). Numbers on this scale are known as magnetite oxidation numbers, and are defined as above with the exception that the total area of grains replaces the total number of grains (n_i or n).

Magnetite granulation, which is very rare in the Canary Islands collection, consists of patches of fine rutile granules within titanomagnetite grains. Measurement

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of this type of alteration depends on the degree of replacement of magnetite by granules. A magnetite granulation number has been defined by Ade-Hall *et al.* (1968) and runs from 1·00 (granulation absent) to 6·00 (complete replacement). Irregularly distributed lightening of the brown colour of typical unaltered titanomagnetite, commonly termed 'maghemitization', is found in a number of Canary Island specimens. Systematic assessment of this feature was discontinued early in the measuring programme in view of the subjectivity involved in judging the degree and extent of this type of alteration both within and between specimens.

Other titanomagnetite properties obtained from the basic measurements were magnetite content and average grain dimension. Magnetite content describes the relationship of total titanomagnetite area to the combined silicate and opaque area per polished section examined. Magnetite content is approximately related to magnetite volume by a (2/3) power law.

Separate ilmenite (as distinct from the ilmenite of deuteric oxidation lamellae in titanomagnetite) and sulphide development are also obtained from the basic measurements. These are all expressed as percentages of the total *opaque* area.

3. Analysis of results

3.1 Cleaned natural remanence directions. These have been subdivided into normal (*N*), anomalous (*A*), and reverse (*R*) polarity groups. The criterion for subdivision was the relationship of the direction of cleaned natural remanence of a specimen to the direction of the magnetic field produced by a geocentric axial dipole at the present mean latitude of the Canary Islands. The mean latitude of the islands, to the nearest degree, was taken as 28° N, and the corresponding geocentric axial dipole field direction for this latitude is (N000E, +47). Specimens were allocated to the *N*, *A* or *R* polarity groups on the basis of the divergence (Blackett 1962) between their cleaned remanence direction and the geocentric axial dipole field direction. Following Dagley *et al.* (1967), a maximum divergence of 40° from the dipole direction is used to define the normal group, a minimum divergence of 140° defines the reverse group while directions with divergence in the 40°–140° range are assigned to the anomalous group.* Experience in Iceland (Dagley *et al.* op. cit.) and elsewhere suggests that at least some of these anomalous directions may represent transitions between the time-dominant normal and reverse states of the geomagnetic field.

As a result of this analysis 85 flows comprise the normal group, 15 flows the anomalous group and 68 flows the reverse group.

For reasons given in Section 4 further division of the data was carried out by subdividing the polarity groups on the basis of silicate petrology. Thus each group was split into ankaramite, basalt and phonolite subgroups using the criteria listed in Section 2(*a*).

3.2 Opaque petrological data. Mean values, together with standard errors of the mean, of petrological parameters have been calculated for each polarity group or subgroup (Tables 1 and 2). In a few cases more than one specimen was examined from a flow. In these cases mean within-flow petrological parameters were first calculated, and these means allowed to contribute to polarity group mean values. That is to say, each flow has been given unit weight in the petrological analysis.

4. Results of analysis

First we consider the whole collection of 168 lavas divided simply into *N*, *A* and *R* polarity groups without consideration of differences in silicate petrology (Table 1). No opaque petrological property shows significant differences between polarity group mean values in this first analysis.

* One of us (NDW) favours a definition based on vertical geomagnetic latitude differences, but this does not significantly affect the analyses made here.

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Table 1

Mean values of opaque petrological parameters for initial division into normal, anomalous and reverse groups

(Errors are standard errors of the mean)

Group	Number of flows	Mean magnetite oxidation number	Mean magnetite grain size (μ)	Mean magnetite content	Mean ilmenite percentage	Mean pyrite percentage	Mean pyrrhotite percentage	Mean combined sulphides percentage
Normal (*N*)	85 or 61*	1·90±0·12 (85)	22±2 (85)	0·0247±0·0029 (61)	12·8±1·9 (85)	0 (61)	0·02±0·01 (61)	0·02±0·01 (61)
Anomalous (*A*)	15 or 10	1·90±0·32 (15)	20±2 (15)	0·0285±0·0062 (10)	10·3±3·0 (15)	0 (10)	0·02±0·10 (10)	0·02±0·10 (10)
Reverse (*R*)	68 or 47	2·03±0·16 (68)	23±2 (68)	0·0259±0·0046 (47)	11·2±2·3 (68)	0 (47)	0·02±0·01 (47)	0·03±0·01 (47)

* Magnetite granulation and content measurements and sulphide measurements were instigated part way through the petrological study.

Table 2

Mean values of opaque petrological parameters for normal anomalous and reverse divisions of silicate petrology based subgroups

(Errors are standard errors of the mean)

Subgroup		Number of flows	Mean magnetite oxidation number	Mean magnetite grain size (μ)	Mean magnetite content	Mean ilmenite percentage	Mean pyrite percentage	Mean pyrrhotite percentage	Mean combined sulphides percentage
Ankaramites	*N*	0	—	—	—	—	—	—	—
	A	0	—	—	—	—	—	—	—
	R	18 or 8	1·95±0·37 (18)	22±2 (18)	0·0539±0·0219 (8)	2·6±0·8 (18)	0·02±0·02 (8)	0 (18)	0·02±0·02 (8)
Basalts	*N*	72 or 48	1·95±0·14 (72)	23±2 (72)	0·0303±0·033 (48)	13·2±2·2 (72)	0 (48)	0·02±0·01 (48)	0·02±0·01 (48)
	A	13 or 8	2·00±0·36 (13)	19±2 (13)	0·0304±0·0076 (8)	10·1±3·3 (13)	0 (8)	0·15±0·11 (8)	0·15±0·12 (8)
	R	47 or 36	1·96±0·18 (47)	21±2 (47)	0·0192±0·0019 (36)	14·4±3·1 (47)	0 (36)	0·03±0·02 (36)	0·03±0·02 (36)
Phonolites	*N*	13	1·72±0·27	12±1	0·0057±0·0011	9·1±2·0	0·01±0·01	0·02±0·02	0·03±0·02
	A	2	1·20±0·00	24±8	0·0211±0·0011	11·8±3·9	0	0	0
	R	3	3·51±0·42	51±30	0·301 ±0·0207	13·0±5·1	0	0·01±0·00	0·01±0·00

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Now, Haggerty (1968) has compared microscopically determined oxidation states of Canary Island lavas with chemically determined whole rock oxidation states, expressed by the ratio $Fe_2O_3/(Fe_2O_3+FeO)$. For basalts Haggerty finds a rather systematic relationship between the two parameters. However, it appears likely that the microscopically determined deuteric oxidation state of phonolites and basalts respectively correspond to different whole rock chemical oxidation state scales. In particular, phonolites with microscopically determined deuteric oxidation states expressed by a magnetite oxidation number of 1·00 (deuteric oxidation entirely absent) have significantly higher $Fe_2O_3/(Fe_2O_3+FeO)$ ratios than basalts with the same magnetite oxidation number. This situation would be consistent either with the primary titanomagnetite of phonolites having a higher magnetite to ulvospinel ratio or a higher degree of cation deficiency than the basaltic titanomagnetite. Thus, the comparison of microscopically determined mean deuteric oxidation states of polarity groups consisting of several types of lava, in different proportions in each group, may obscure the presence of oxidation state differences between polarity groups of lavas of the same composition. To avoid the possibility of correlations between polarity and opaque petrology being obscured in this way the polarity groups were subdivided on a basis of silicate petrology into ankaramitic, basaltic and phonolitic groups. Comparison of *N*, *A* and *R* mean values of opaque petrological parameters for these subgroups is seen in Table 2.

The small group of ankaramitic lavas (18 flows) is of interest in that all the flows are reversely magnetized. This is an unexpected result which in itself deserves attention. It seems possible that the explanation of this result is that most of the ankaramites belong to a single volcanic group on Teneriefe (on the Ladera de Guimar Scarp) which Abdel-Monem, Gast & Watkins (1967) have shown to be extruded during the Matuyama reversed polarity epoch. Thus, the association of the ankaramites uniquely with reserve polarity is apparently of low significance.

The basalts, with 132 flows, form the largest silicate petrological group. Here a significant difference between polarity divisions only exists for magnetite content, with the *R* group mean value significantly lower than the indistinguishable *N* and *A* mean values.

The small group of phonolites, with 18 flows, has two interesting characteristics. First, 13 of the flows are normally magnetized (the two anomalous flows also have directions of magnetization close to the normal group limit). The explanation of this preponderance of normal flows appears to be the same as that suggested for the association of ankaramites and reverse polarity, that is, most of the phonolites are from the bodies extruded during the Brunhes normal polarity epoch on Teide in Teneriefe. Significant differences in magnetite oxidation number, grain size and content exist between the phonolite *N*, *A* and *R* divisions, with the *R* divisions having higher values for each parameter and the *A* values being intermediate in the cases of grain size and content. However, the numbers of *A* and *R* flows (2 and 3 respectively) are too small for these differences to be considered seriously.

Thus, with the possible exception of the case of magnetite content in the basaltic lava group, at this new level of subdivision of the data there still appears to be no significant relationship between opaque petrology and natural remanence polarity in these Canary Island lavas.

A second way in which a polarity–petrology correlation might be obscured concerns the problem of low deuteric oxidation specimens with associated single (above ambient temperature) high ($\geqslant$ 500 °C) strong field Curie points. Specimens having this association of apparent *low* deuteric oxidation and single *high* strong field Curie points have been found in several other collections (Ade-Hall *et al.* 1965). Their relevance to the results being discussed stems from the possibility that these high single Curie points may indicate the presence of considerable deuteric oxidation, perhaps involving ilmenite lamellae too fine to be observed with an ore microscope.

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Evidence for the existence of such fine exsolution, where lamellae could only be resolved with a scanning electron microscope, has been described by Wasilewski (1969). Wasilewski's finding probably does not hold generally since variation of strong field magnetization to liquid nitrogen temperatures generally failed to show a below ambient temperature ilmenite Curie point (Ade-Hall *et al.* 1965).

However, let us suppose for the moment that Wasilewski's explanation holds true for the Canary Islands collection, and follow the consequence for petrology–polarity relationships. If the majority of low deuteric oxidation, single high Curie point specimens were reversely magnetized, this would imply, so far as magnetite deuteric oxidation was concerned, that deuteric oxidation was preferentially associated with reverse polarity. Conversely, if the specimens in question were normally magnetized, high deuteric oxidation would be associated with normal polarity.

Strong field heating curves are available for 53 specimens of which 12 fall into the low deuteric oxidation, single high Curie point category. Of these 12 specimens five are of normal polarity and five of reverse polarity, the remaining two being of anomalous polarity. It seems unlikely that any significant differences between N and R group mean values of magnetite deuteric oxidation could result from this symmetrical distribution of low deuteric oxidation, single high Curie point specimens.

5. Discussion

The general picture concerning the relationship between petrology and natural remanence polarity in basaltic rocks has yet to be satisfactorily defined. A number of examples of significant associations of relatively high magnetite deuteric oxidation and/or ilmenite percentage with reverse polarity have been reported (for example: Wilson 1966; Wilson & Watkins 1967; Watkins & Haggerty 1968; Ade-Hall & Wilson, 1969). Other collections show no significant correlations (Larson & Strangway 1966, Watkins & Haggerty 1968). The Canary Islands collection belongs to this last category. Several ways in which correlations could have been obscured have been considered for this collection without the detection of any correlations. No collection has yet been reported as showing correlations of high deuteric oxidation or ilmenite percentage with normal polarity.

Unfortunately, the slow rate at which quantitative petrological studies can be carried out has resulted in the production of only a small number of well-documented examples. Thus it is not feasible yet to look for such features as age, geographical or geological environment restrictions on correlations.

Two suggestions have been made which can be tested using the existing information. The first, due to Strangway (personal communication), is that correlations of trivial significance may occur when rocks adjacent to a *single* polarity transition are examined. Two properties of known correlations are incompatible with this suggestion. First, in several cases collections from one area include material from many polarity intervals. The example of the correlation of relatively high ilmenite percentage with reverse polarity in Scotland (Carboniferous) lavas (Wilson 1966) illustrates this point (Fig. 1). Secondly, an equal number of correlations of, say, relatively high magnetite deuteric oxidation with normal and reverse polarity might be expected to be found if this suggestion was correct. In fact, correlations, where they exist, so far always favour reverse polarity.

Secondly, Hide (1966) has proposed that geomagnetic field reversals consist of two types, forced and free, and that petrological changes might be expected to correlate with the longer period forced reversals. Now, several of the reported correlations (Wilson 1966; Watkins & Haggerty 1968; Ade-Hall & Wilson 1969) may relate to basaltic rocks intruded or extruded during periods characterized by high reversal frequency, that is, reversals probably of the free type. Thus correlations

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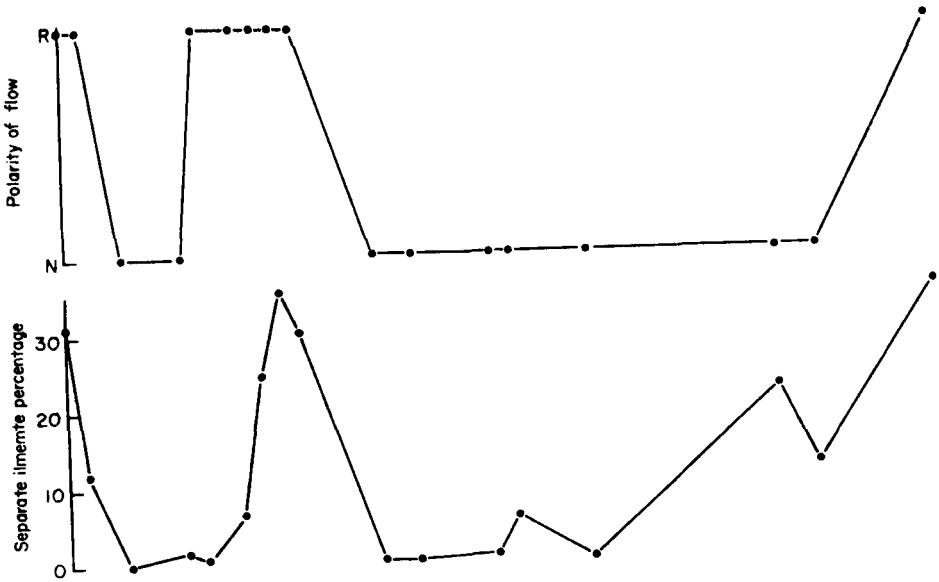


FIG. 1. Variation of separate ilmenite content with polarity in a sequence of Scottish Carboniferous lavas (Wilson 1966) in a series of lava flows and interbedded sediments. (Original stratigraphic column after Geikie (1900).)

sometimes can occur during periods of frequent field reversals. It remains to be seen whether they are more clear cut during periods of low reversal frequency.

Further work seems necessary to assess the meaning and importance of relationships between petrology and natural remanence polarity. Notwithstanding the fact that the Canary Islands result reported here shows no correlations between petrology and natural remanence polarity, we are pursuing two other lines. Following the suggestion of Bullard (1968) correlations are being searched for in the basaltic lavas that contribute to the polarity-time scale of Cox, Doell & Dalrymple (1968) for the last four million years and, as suggested by Wright (1968), in the sediments of deep sea cores.

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